

Statistical Analysis Questions and Answers

Q1. What is bias in statistics? Discuss different types of bias with examples.

1. Meaning of bias in statistics

- Bias is a systematic error that causes an estimator, sample, or study result to consistently deviate from the true value of the population parameter.
- It arises from the way data are collected, measured, or modeled, and leads to distorted or unrepresentative conclusions.

2. Selection bias

- Occurs when the sample is not representative of the target population because of the way participants are selected.
- Example: Testing a new medicine only on young, healthy men and generalizing the result to all adults, including women and elderly patients.

3. Sampling bias / self-selection bias

- Sampling bias appears when some members of the population are more likely to be included than others, often due to non-random sampling.
- Example: An online survey about internet usage that only reaches people with good internet access will over-represent heavy internet users.

4. Measurement / observer bias

- Arises when the way data are measured or recorded systematically favors certain outcomes.
- Example: A doctor who strongly believes a drug works may unconsciously rate symptoms as "improved" more often in the treatment group (observer bias).

5. Recall bias

- Happens when participants do not remember past events accurately, and the accuracy differs between groups.
- Example: In a cancer study, patients with the disease may recall their past exposures (e.g., smoking) more carefully than healthy controls.

6. Survivorship bias

- Occurs when only "survivors" or successful cases are analyzed, and failures are ignored.
- Example: Studying only successful start-ups to learn how to build a company and ignoring thousands of failed start-ups leads to over-optimistic conclusions.

7. Confirmation / cognitive bias in analysis

- Analysts tend to search for, interpret, and report results that confirm their prior beliefs, ignoring conflicting evidence.
- Example: During elections, people share only polls that favor their preferred candidate, giving a biased view of actual support.

Q2. What is sampling? Discuss different types of sampling methods and the importance of sample size.

1. Definition of sampling

- Sampling is the process of selecting a subset (sample) of individuals from a larger group (population) to draw conclusions about the whole population.
- It is used when studying every member of the population (census) is too costly, slow, or impossible.

2. Probability sampling methods (random selection, known probabilities)

- **Simple random sampling:** Every member has equal chance; e.g., lottery or random number table.
- **Systematic sampling:** Select every k -th unit after a random start (e.g., every 10th customer).
- **Stratified sampling:** Population is divided into strata (e.g., male/female, regions) and random samples are taken from each stratum to ensure representation.
- **Cluster sampling:** Population is divided into clusters (e.g., schools, villages); some clusters are randomly selected, and all or some units within them are studied.

3. Non-probability sampling methods (non-random, unknown selection probabilities)

- **Convenience sampling:** Take whoever is easily available (e.g., students in your class); cheap but highly biased.
- **Purposive (judgmental) sampling:** Researcher selects "information-rich" cases based on expert judgment.
- **Snowball sampling:** Existing participants recruit others (common in hard-to-reach populations).
- **Quota sampling:** Select units to fill pre-defined quotas (e.g., 50% male, 50% female) without random selection.

4. Importance of sample size

- Larger samples reduce sampling error and give more precise estimates of population parameters.
- Too small a sample can lead to low statistical power and failure to detect true effects; too large a sample can be wasteful and may detect trivial differences that are not practically important.

Q3. Explain the misuse and misinterpretation of statistics. Give two real-life examples.

1. What is misuse of statistics?

- Misuse occurs when statistical methods, numbers, or graphs are used in a way that leads to unjustified or incorrect conclusions, either intentionally or by ignorance.
- This includes wrong methods, biased data, misleading averages, cherry-picking results, and deceptive visualizations.

2. Common ways statistics are misused

- Using very small or biased samples and generalizing to the whole population.
- Reporting only favorable results (cherry-picking) and hiding negative findings.
- Using misleading averages (e.g., mean instead of median when data are highly skewed) to present a better picture.
- Drawing graphs with distorted scales to exaggerate or hide differences.

3. Example 1 - Misleading average salary

- Universities and companies sometimes quote the "average" salary of graduates or employees using the mean, which is pulled up by a few very high earners.
- Most individuals may actually earn much less than this mean, so the statistic gives a misleading impression of typical earnings.

4. Example 2 - Biased election polls

- The famous 1936 *Literary Digest* poll predicted the wrong winner of the U.S. presidential election because their survey mainly reached wealthier people (telephone and car owners), not a representative cross-section of voters.
- This biased sample led to an incorrect conclusion that Alfred Landon would win, while Franklin Roosevelt actually won by a

landslide.

Q4. Explain the process of data preparation and data cleaning in statistical analysis.

1. Data preparation (overall process)

- Data preparation converts raw data into a clean, structured form suitable for analysis by integrating, cleaning, transforming, and reducing data.
- Typical steps: data integration, data cleaning, data transformation (scaling/encoding), data reduction, and sometimes sampling or discretization.

2. Data cleaning steps

- **Identify problems:** Detect missing values, outliers, duplicates, inconsistent formats, and impossible values using summaries and plots (histograms, boxplots, IQR, etc.).
- **Remove duplicates and irrelevant records:** Drop repeated rows or records that are outside the scope of the study.
- **Handle missing values:** Options include deletion (if few and random) or imputation (mean, median, mode, or model-based) depending on the situation.
- **Fix structural errors:** Standardize formats (e.g., "M/F" vs "Male/Female"), correct typos, unify units (kg vs g), and ensure correct data types.

3. Data transformation and reduction

- **Scaling and normalization:** Apply methods such as standardization (z-score), min-max scaling, or robust scaling so that features are on comparable scales, which is important for many models.
- **Encoding categorical variables:** Convert categories into numeric form (one-hot encoding, label encoding) so that algorithms can process them.
- **Feature selection/reduction:** Remove redundant or irrelevant attributes to simplify models, improve performance, and reduce overfitting.

4. Validation of cleaned data

- After cleaning and preparation, results are validated using cross-validation, hold-out datasets, or comparison with external benchmarks to ensure data quality and stability of conclusions.
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Q5. Illustrate the concept of statistical error with examples from data analysis.

1. Meaning of statistical error

- A statistical error is the difference between an observed sample result and the true (but unknown) population value due to randomness, measurement, or model assumptions.
- Errors occur even when there is no mistake; they reflect uncertainty inherent in sampling and inference.

2. Sampling error

- Arises because we observe only a sample, not the entire population, so sample statistics (mean, proportion) differ from population parameters.
- Example: The average height in a sample of 50 students will not be exactly equal to the true average height of all students in the college, even if the sample is random.

3. Type I and Type II errors in hypothesis testing

- **Type I error (false positive):** Rejecting a true null hypothesis, e.g., concluding a new drug works when in reality it does not.
- **Type II error (false negative):** Failing to reject a false null hypothesis, e.g., concluding there is no difference between two treatments when one is actually better.

4. Measurement and model errors

- **Measurement error:** Difference between the true value and the measured value because of instrument inaccuracy or human mistakes (e.g., mis-keying survey responses).

- **Model error (residuals):** In regression, residuals represent the difference between observed values and predicted values; they capture unexplained variation and model misfit.
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Q6. Explain how statistical modeling and computational statistics are used in solving real-world problems.

1. Statistical modeling - concept

- A statistical model is a mathematical description of how data are generated, typically expressed as probability distributions and relationships between variables.
- Models like regression, classification, clustering, and time-series analysis help us understand relationships, quantify uncertainty, and make predictions.

2. Real-world applications of statistical modeling

- **Healthcare:** Logistic regression and time-series models predict patient readmission risk and forecast disease spread to allocate resources.
- **Finance:** Classification and regression models estimate credit risk, detect fraud, and forecast asset prices.
- **Marketing:** Models segment customers, predict churn, and optimize pricing and promotions based on historical purchase data.

3. Computational statistics

- Computational statistics uses algorithms and computing power (simulation, resampling, optimization) to fit complex models that are analytically intractable.
- Techniques such as bootstrap, Monte Carlo simulation, Bayesian computation (MCMC), and high-dimensional optimization allow analysts to estimate uncertainty and model complex systems.

4. Benefits in problem-solving

- These approaches support evidence-based decision-making, allowing organizations to test scenarios, evaluate policies, and deploy predictive systems in real time.
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Q7. Analyze Type I and Type II errors of hypothesis testing using a suitable example.

1. Definitions

- **Type I error (α):** Rejecting the null hypothesis when it is actually true (false positive).
- **Type II error (β):** Failing to reject the null hypothesis when it is actually false (false negative).

2. Example: Medical diagnostic test

- Null hypothesis H_0 : "The patient does not have the disease."
- Alternative hypothesis H_1 : "The patient has the disease."
- **Type I error:** Test says "disease present" when the patient is actually healthy -- leads to unnecessary anxiety and treatment.
- **Type II error:** Test says "no disease" when the patient is actually ill -- may delay treatment and cause serious harm.

3. Control of errors

- The significance level α (often 0.05) is the probability of making a Type I error; lowering α reduces false positives but may increase β .
 - Statistical power $1 - \beta$ is the probability of correctly rejecting a false null hypothesis; higher power reduces Type II errors and depends on sample size, effect size, and variability.
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Q8. Explain the terms degrees of freedom, power, robustness and Non-parametric analysis in statistical testing.

1. Degrees of freedom (DF)

- DF is the number of independent values that can vary in the calculation of a statistic after accounting for constraints.
- Often approximated as sample size minus the number of estimated parameters; for example, estimating variance from N observations uses $N - 1$ degrees of freedom.

2. Power of a test

- Power is the probability that a statistical test correctly rejects a false null hypothesis (detects a real effect).
- It equals $1 - \beta$, where β is the probability of a Type II error; higher power is achieved with larger samples, stronger effects, and appropriate test choices.

3. Robustness

- A procedure is robust if its performance is not highly sensitive to violations of assumptions or to outliers.
- Robust statistics include the median and interquartile range, which are less affected by extreme values than the mean and range.

4. Non-parametric analysis

- Non-parametric tests do not assume a specific parametric distribution (like normal) for the data and often work with ranks or signs instead of raw values.
 - They are particularly useful when data are skewed, ordinal, or contain outliers, and can be more powerful than parametric tests when normality is violated.
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Q9. A researcher collected a sample of 36 observations with a sample mean of 72 and a standard deviation of 12. Construct the 95% confidence interval for the population mean. (Use $z = 1.96$)

1. Given values

- Sample size: $n = 36$
- Sample mean: $\bar{x} = 72$
- Sample standard deviation: $s = 12$
- For 95% CI, $z = 1.96$ (standard normal critical value).

2. Standard error of the mean (SEM)

- $$\text{SEM} = \frac{s}{\sqrt{n}} = \frac{12}{\sqrt{36}} = \frac{12}{6} = 2.$$

3. Margin of error (ME)

- $$\text{ME} = z \times \text{SEM} = 1.96 \times 2 = 3.92.$$

4. 95% confidence interval

- Lower limit: $\bar{x} - \text{ME} = 72 - 3.92 = 68.08.$
- Upper limit: $\bar{x} + \text{ME} = 72 + 3.92 = 75.92.$
- So, the 95% CI for the population mean is (68.08, 75.92).

Q10. A teacher recorded the number of books read by 12 students in a month

Data: 2, 5, 3, 5, 7, 2, 9, 5, 3, 7, 2, 6

- a) Count: Find the total number of observations in the dataset.
- b) Variety: List all the distinct (unique) values present in the dataset.
- c) Calculate the total sum of all observations.
- d) Find the average number of books read.
- e) Calculate the Majority and Minority values.
- f) Find the minimum and Maximum values in a given dataset.

1. (a) Count (number of observations)

- There are 12 values, so the count is **12**.

2. (b) Variety (distinct values)

- Unique values in the dataset: **2, 3, 5, 6, 7, 9**.

3. (c) Total sum of observations

- Sum = $2 + 5 + 3 + 5 + 7 + 2 + 9 + 5 + 3 + 7 + 2 + 6 = 56$.
- Total sum = **56**.

4. (d) Average (mean) number of books

- Mean = $\frac{\text{sum}}{n} = \frac{56}{12} \approx 4.67$ books.

5. (e) Majority and minority values (modes / least frequent)

- Frequency counts:
 - 2 → 3 times; 3 → 2 times; 5 → 3 times; 6 → 1; 7 → 2; 9 → 1.
- **Majority values (modes):** 2 and 5 (highest frequency 3).
- **Minority values:** 6 and 9 (lowest frequency 1).

6. (f) Minimum and maximum

- Minimum value = **2**.
- Maximum value = **9**.

Q11. Explain Measures of Spread (Range, Variance, SD, IQR) with formulas and examples. Find the Variance and Standard Deviation: 5, 7, 9, 11, 13.

1. Range

- Definition: Difference between maximum and minimum values.
- Formula: $\text{Range} = \max(x) - \min(x)$.
- It is easy to compute but very sensitive to outliers.

2. Variance and Standard Deviation

- Variance measures the average squared deviation of values from the mean.
- For a sample of size n with mean $\bar{x}$:
Sample variance $s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$.
- Standard deviation (SD) is the square root of variance and represents the typical distance of observations from the mean.

3. Interquartile Range (IQR)

- IQR is the difference between third quartile Q_3 and first quartile Q_1 : $\text{IQR} = Q_3 - Q_1$.
- It measures spread of the middle 50% of the data and is robust to outliers.

4. Variance and SD for data 5, 7, 9, 11, 13

- Data: 5, 7, 9, 11, 13; $n = 5$.
 - Mean: $\bar{x} = (5 + 7 + 9 + 11 + 13)/5 = 45/5 = 9$.
 - Deviations: $-4, -2, 0, 2, 4$.
 - Squared deviations: 16, 4, 0, 4, 16; sum = 40.
 - **If treated as a sample:**
 - Variance $s^2 = 40/(5 - 1) = 40/4 = 10$.
 - SD $s = \sqrt{10} \approx 3.16$.
 - **If treated as a population:**
 - Variance $\sigma^2 = 40/5 = 8$.
 - SD $\sigma = \sqrt{8} \approx 2.83$.
 - In most data-analysis contexts these numbers would be considered a sample, so $s^2 = 10$ and $s \approx 3.16$ are usually reported.
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Q12. A dataset representing marks of students is given below

Data: 12, 15, 18, 20, 22, 25, 28, 30, 35, 40

Construct a box plot for the given data and calculate:

- a) Minimum
 - b) Maximum
 - c) Calculate Q_1 , Median, and Q_3
 - d) Compute the IQR
 - e) Identify any outliers using the $1.5 \times \text{IQR}$ rule
- Draw the modified box plot

1. (a) Minimum and (b) Maximum

- Minimum = **12**.
- Maximum = **40**.

2. (c) Quartiles and median

- Median (Q_2): Average of 5th and 6th values = $(22 + 25)/2 = 23.5$.
- Lower half: 12, 15, 18, 20, 22 $\rightarrow Q_1$ is the median of these = **18**.
- Upper half: 25, 28, 30, 35, 40 $\rightarrow Q_3$ is the median of these = **30**.

3. (d) Interquartile range (IQR)

- $\text{IQR} = Q_3 - Q_1 = 30 - 18 = 12$.

4. (e) Outliers using $1.5 \times \text{IQR}$ rule

- Lower fence = $Q_1 - 1.5 \times \text{IQR} = 18 - 1.5 \times 12 = 18 - 18 = 0$.
- Upper fence = $Q_3 + 1.5 \times \text{IQR} = 30 + 18 = 48$.
- Any data < 0 or > 48 is an outlier; all marks (12–40) lie within $[0, 48]$, so **there are no outliers**.

5. Modified box plot (description)

- Draw a box from $Q_1 = 18$ to $Q_3 = 30$ with a line at the median 23.5.
 - Whiskers extend from the box to the minimum 12 and maximum 40.
 - No separate outlier symbols because no values fall outside the fences.
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Q13. Describe Measures of Distribution Shape Skewness and Kurtosis with diagrams.

1. Skewness - asymmetry of the distribution

- Skewness measures how symmetric or asymmetric a distribution is around its mean.
- **Positive (right) skew:** Long right tail; mean > median; e.g., income distributions where a few very rich people pull the mean to the right.
- **Negative (left) skew:** Long left tail; mean < median; e.g., test scores where most students score high but a few get very low marks.

2. Kurtosis - peakedness and tail thickness

- Kurtosis describes how heavy the tails are and how peaked or flat the distribution is compared with the normal distribution.
- Types:
 - **Leptokurtic:** Sharper peak and fatter tails than normal (many extreme values).
 - **Mesokurtic:** Similar to normal distribution (reference case).
 - **Platykurtic:** Flatter peak and thinner tails; fewer extreme values than normal.

3. Use of moments

- Skewness and kurtosis are based on the third and fourth central moments of the distribution, scaled by powers of the standard deviation to make them dimensionless.
 - Together with mean and variance (first two moments), they provide a fuller description of the shape of a distribution.
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Q14. Describe the role of statistical indices in summarizing data.

1. What are statistical indices?

- Index numbers are statistical measures that express changes in one or more related variables (prices, quantities, values) relative to a base period, typically set to 100.
- Examples: Consumer Price Index (CPI), stock market indices (S&P 500), production indices, wage indices.

2. Summarizing complex data

- Indices condense large volumes of data into a single percentage figure, making it easier to understand overall trends without tracking each individual item.
- For example, a CPI of 120 indicates that the price level has increased by 20% compared to the base period.

3. Comparisons over time and space

- They allow comparison of changes across time (e.g., inflation over years) and across regions (cost-of-living indices for different cities).
- Composite indices combine many series into one measure, supporting macro-level decisions in policy, economics, and business.

4. Decision-making and policy use

- Governments use price and wage indices to adjust salaries, pensions, and tax brackets for inflation.
 - Businesses track production and market indices to forecast demand and plan investment.
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Q15. Explain the importance of moments in describing a statistical distribution.

1. What are moments?

- Moments are numerical measures derived from powers of deviations of data values from a reference point (often the mean or zero), summarizing different aspects of a distribution.
- There are raw moments (about origin) and central moments (about the mean). Central moments are widely used in statistics.

2. Key moments and what they describe

- **First central moment:** Always zero by definition.
- **Second central moment (variance):** Measures dispersion (spread) of data around the mean.
- **Third central moment (skewness):** Captures asymmetry of the distribution.
- **Fourth central moment (kurtosis):** Captures tail heaviness and peakedness.

3. Why moments are important

- Moments provide a systematic way to describe and compare distributions beyond mean and variance alone.
 - Many statistical methods (e.g., normality tests, approximations, advanced models) rely on estimates of moments to understand underlying distributional properties.
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Q16. Explain statistical key functions to summarize a dataset.

1. Measures of central tendency

- **Mean:** Arithmetic average; sensitive to outliers.
- **Median:** Middle value when data are ordered; robust to outliers, preferred for skewed distributions.
- **Mode:** Most frequent value, useful for categorical or discrete data.

2. Measures of spread (variability)

- **Range:** Max – Min; simplest measure of spread.
- **Variance and standard deviation:** Average squared deviation and its square root, describing how spread out data are around the mean.
- **Interquartile range (IQR):** Spread of the middle 50% of data; robust to outliers.

3. Position and shape measures

- **Percentiles and quartiles:** Indicate the relative standing of a value within the dataset (e.g., 90th percentile).
- **Skewness and kurtosis:** Summarize asymmetry and tail behavior, complementing basic summaries.

4. Counting and aggregation functions (as in tools like Excel, R, Python)

- **COUNT:** Number of observations.
 - **SUM:** Total of all values.
 - **MIN / MAX:** Smallest and largest values.
 - **DESCRIPTIVE STATISTICS packages:** Provide mean, median, SD, IQR, etc., in one step, giving a quick summary profile of a dataset.
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Q17. Compare different measures of model complexity and examine how they influence the model selection process.

1. Simple measures of model complexity

- **Number of parameters / features:** More parameters usually mean a more complex model that can fit data more flexibly but risks overfitting.
- **Model structure:** Deep neural networks, high-degree polynomials, and high-order ARIMA models are more complex than linear regression with few predictors.

2. Information-criteria-based measures - AIC and BIC

- **AIC (Akaike Information Criterion):** Balances goodness of fit (likelihood) with a penalty proportional to the number of parameters; lower AIC indicates a better trade-off for prediction-oriented tasks.
- **BIC (Bayesian Information Criterion):** Similar to AIC but with a stronger penalty that grows with sample size, tending to select simpler, more parsimonious models.

3. Other complexity-aware measures

- **Adjusted R^2 :** Modifies R^2 by penalizing unnecessary predictors so that adding a variable only increases adjusted R^2 if it truly improves fit.
- **Cross-validation error:** Estimates out-of-sample predictive performance, effectively punishing models that overfit training data even if in-sample error is low.

4. Influence on model selection

- **Trade-off between bias and variance:** Simple models may underfit (high bias), while very complex models may overfit (high variance); good criteria help find a balance.
- In practice, analysts compare candidate models using AIC/BIC, adjusted R^2 , and cross-validation, then choose a model that has

low complexity but still achieves strong predictive or explanatory performance.